



An Improved Deep Neural Network System for Strawberry Ripeness Detection and Classification

Ömer Özcan¹ · Muhammed Karaaltun² · Mehmet Hacıbeyoğlu²

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Abstract

Accurate identification of fruit ripeness is crucial for ensuring product quality, optimizing harvest timing, and minimizing postharvest losses in modern agriculture. Among various fruit crops, strawberries present a particular challenge due to their high morphological variability, delicate structure, and short shelf life, making automated, high-precision inspection essential for modern agricultural systems. To address these challenges, this study proposes an improved whale optimization algorithm (IWOA)-based strawberry detection and ripeness inspection (S-Det) system that integrates optimization-driven feature extraction with deep learning-based classification. The proposed framework combines an IWOA-optimized autoencoder for compact and discriminative feature representation with an IWOA-optimized deep neural network (DNN) for precise multi-class classification of strawberry ripeness stages. Two key improvements are introduced into the traditional whale optimization algorithm (WOA): population diversity monitoring using cosine similarity and an improved spiral position updating mechanism, which together enhance convergence speed and prevent premature stagnation in complex agricultural datasets. Comprehensive experiments were conducted on the StrawTex dataset, consisting of 6000 strawberry texture samples across five ripeness categories. The proposed IWOA-DNN achieved a classification accuracy of 98.23%, precision of 98.25%, recall of 98.26%, and an F1-score of 0.9824, outperforming classical models such as *k*-nearest neighbors (KNN, 92.17%) and support vector machine (SVM, 92.06%), as well as advanced methods like particle swarm optimization–DNN (PSO-DNN, 97.62%), WOA-DNN (97.72%), and pre-trained architectures including ResNet50 (97.73%). These findings confirm the superior learning ability, convergence stability, and generalization capability of the IWOA-based system, supporting its potential deployment in real-world agricultural automation. Overall, the results demonstrate that the proposed S-Det framework provides an effective, intelligent, and highly reliable solution for strawberry ripeness detection, offering strong potential for real-time implementation in smart sorting lines and robotic harvesting systems. Furthermore, the proposed approach shows promise for broader applications across various precision agriculture tasks.

Keywords Precision agriculture · Metaheuristic optimization · Autoencoder-based feature learning · Texture analysis · Agricultural computer vision

Introduction

Strawberry (*Fragaria × ananassa*) is one of the most economically valuable fruit crops globally, providing significant contributions to agricultural income, employment, and international trade. According to the Food and Agriculture Organization (FAO), global strawberry production exceeds 9 million tons annually, with a continuous upward trend due to high consumer demand and its extensive use in the food processing industry. The strawberry sector generates billions of dollars each year and supports rural economies through intensive labor requirements during planting, harvesting, and postharvest handling. However, despite its high

✉ Ömer Özcan
oozcan@selcuk.edu.tr

Muhammed Karaaltun
muhammed.karaaltun@erbakan.edu.tr

Mehmet Hacıbeyoğlu
hacibeyoglu@erbakan.edu.tr

¹ Department of Information Technologies, Selcuk University, 42060 Selcuklu, Konya, Turkey

² Department of Computer Engineering, Faculty of Engineering, Necmettin Erbakan University, 42090 Meram, Konya, Turkey

economic importance, strawberries are delicate fruits with short shelf lives and high harvesting costs (Delbridge 2021).

Accurately determining the ripeness level of strawberries is crucial for maintaining fruit quality, optimizing harvesting time, and minimizing postharvest losses. Overripe fruits reduce shelf life and exportability, while underripe fruits decrease consumer satisfaction and market value. Therefore, the automated detection of strawberry ripeness levels plays a vital role in precision agriculture and modern horticultural production systems. Intelligent image-based systems that can identify the visual characteristics of ripeness—such as color, texture, and surface gloss—enable farmers to make timely harvesting decisions, improve grading accuracy, and enhance profitability.

The agriculture industry increasingly employs modern technology to save costs and boost productivity (Mahmud et al. 2023). Research efforts are ongoing across various stages of the agricultural process, including soil cultivation, planting, harvesting, disease detection, and crop analysis (Akumbom et al. 2023). For example, Li et al. (Li et al. 2023) optimized soil tillage methods for wheat cultivation in dry regions using mathematical optimization algorithms to conserve soil moisture and increase yield. Tufail et al. (2020) designed an autonomous planting robot equipped with computer vision to calculate and execute precise planting parameters, while Fan et al. (2022) developed a system for autonomous monitoring of crop growth stages using image processing techniques. Disease diagnosis is another major application area: Srikanth et al. (2023) proposed a mango leaf disease detection model with pesticide recommendation, and Thilagavathi et al. (2023) successfully distinguished healthy and diseased oranges using machine learning with 90% accuracy.

In industrial-scale agriculture, fruits are typically harvested once they reach a specific ripeness threshold to ensure optimal quality. Yoshida et al. (2022b) developed a dual-arm robot capable of harvesting fruits using computer vision, while Amisha et al. (2023; Çeçen 2025) introduced an energy-efficient solar-powered harvesting robot. One of the greatest challenges in agricultural robotics remains environmental perception through computer vision systems (Yoshida et al. 2022a). Focusing on strawberries, Pérez-Borrero et al. (2020) enhanced the Mask region-based convolutional neural network (Mask R-CNN) framework to detect strawberries in natural scenes, while Yoshida et al. (2022a) used depth-sensing cameras to determine 3D fruit positions.

The motivation behind this study stems from the growing need for highly reliable and computationally efficient visual perception modules in agricultural robotics. Although deep neural networks (DNNs) offer outstanding accuracy, their training typically relies on gradient-based optimizers that may become trapped in local minima, leading to subopti-

mal convergence and limited generalization under complex field conditions. To overcome these challenges, this study proposes a novel strawberry detection and ripeness inspection (S-Det) system based on the improved whale optimization algorithm (IWOA). The proposed S-Det framework integrates IWOA-optimized autoencoder-based feature extraction and IWOA-optimized DNN-based classification to jointly enhance the detection and classification of strawberry ripeness levels. The primary contribution of this research is the development of a robust, optimization-driven deep learning framework that ensures stable convergence and superior accuracy. Comparative experiments with state-of-the-art algorithms validate the superiority of the proposed method in achieving high precision, robustness, and real-time applicability in agricultural automation systems.

Materials and Methods

Strawberry Dataset

The Strawberry Digital Images (StrawDI) dataset, which consists of 8000 high-resolution color images captured from natural strawberry fields, was used to construct the Strawberry Textures (StrawTex) dataset (Pérez-Borrero et al. 2020). These images reflect realistic agricultural conditions, including variations in lighting, background, and fruit orientation, making them highly suitable for developing and testing vision-based agricultural models. From the StrawDI dataset, the StrawTex dataset was generated by segmenting the strawberry images into smaller texture patches to emphasize local visual characteristics such as surface color, gloss, and pattern distribution. The StrawTex dataset includes five distinct classes representing the primary ripeness stages of strawberries: white, green, turning, ripe, and background regions (Ozcan 2024). Each class contains images with a resolution of 32×32 pixels, en-

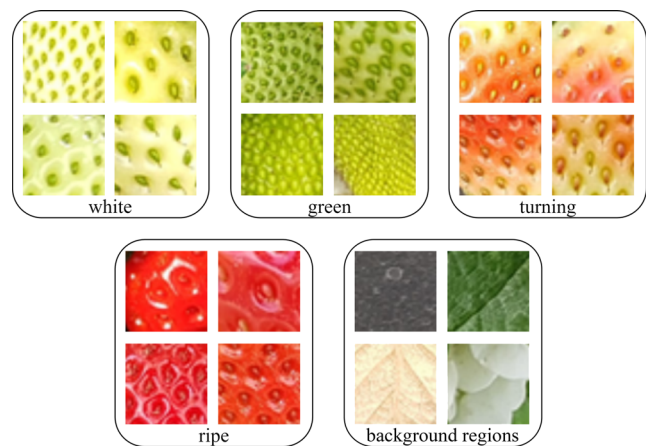


Fig. 1 Example images from the StrawTex dataset

asuring consistent texture-level input for machine learning analysis (Wang et al. 2020). The resulting StrawTex dataset comprises a total of 2000 texture samples, with 400 samples per class, and was used to train the DNN model for accurate strawberry detection and ripeness classification. Figure 1 presents representative examples from each class, illustrating the visual diversity across different ripeness levels. In natural environments, strawberry plants exhibit substantial morphological variability in their fruits, branches, flowers, and leaves, particularly in terms of direction, color intensity, and size (Pérez-Borrero et al. 2020). This diversity poses a challenge for artificial intelligence (AI)-based classification models, which require a sufficiently large and diverse dataset to effectively learn such variations (Min et al. 2023). To address this challenge and enhance the learning capacity of the DNN, data augmentation techniques such as rotation, translation, and zooming were applied to augment data variability and enhance generalization to the StrawTex dataset. These augmentation processes increased the total number of texture samples from 2000 to 6000, enabling the network to better generalize across diverse field conditions and reducing the risk of overfitting. The augmented dataset thus provides a more robust foundation for accurate and scalable strawberry ripeness detection in real-world agricultural scenarios.

Whale Optimization Algorithm and Its Improvement

Whale Optimization Algorithm

The whale optimization algorithm (WOA) is a metaheuristic optimization method inspired by the bubble-net hunting behavior of humpback whales in the oceans (Mirjalili and Lewis 2016). The algorithm simulates the natural hunting strategy of whales to locate and trap prey and is widely used to solve continuous optimization problems.

WOA operates through three main phases:

- Exploration (searching for prey): search agents, representing whales, start from random positions and explore the search space to locate potential prey (solutions).
- Encircling the prey: once the best solution is identified, other whales move toward this optimal solution by updating their positions according to the distance and attraction coefficients.
- Exploitation (bubble-net attacking): whales employ a logarithmic spiral trajectory to simulate the bubble-net feeding process, gradually converging toward the optimal solution.

Although WOA provides a good balance between exploration and exploitation, it still suffers from certain drawbacks, including premature convergence, loss of population

diversity, and slow convergence when dealing with complex or high-dimensional search spaces.

Improved Whale Optimization Algorithm

Meta-heuristic optimization algorithms operate in two fundamental modes: exploration and exploitation.

Exploration allows the algorithm to search broadly across the solution space to discover promising regions, whereas exploitation focuses on intensively refining the search around the best solutions found so far.

In the standard WOA, the balance between these two modes may degrade during later iterations, leading to reduced population diversity and a higher risk of premature convergence. To address these limitations, an IWOA is developed in this study. The IWOA introduces two main enhancements designed to improve both global search ability and local convergence performance.

a) Population Diversity Monitoring

To maintain diversity among search agents, IWOA monitors the similarity between the best solution and other agents using the cosine similarity metric, as expressed in Eq. 1:

$$\text{cosine}(\vec{M}, \vec{N}) = \frac{\vec{M} * \vec{N}}{\|\vec{M}\| * \|\vec{N}\|} \quad (1)$$

where $\vec{M}$ represents the vector of the best agent and $\vec{N}$ represents the vector of another agent in the population.

When the cosine similarity between agents exceeds a threshold ($\beta=0.9$), it indicates a reduction in population diversity and a risk of stagnation in local optima. At this stage, a diversity enhancement mechanism is triggered to guide the search agents toward more promising regions of the solution space.

b) Improved Spiral Position Updating Mechanism

In addition to diversity monitoring, an improved spiral position updating mechanism (ISPUM) is introduced to enhance the exploitation phase. In the standard WOA, whales follow a symmetric logarithmic spiral trajectory toward the prey. In the IWOA, a sinusoidal term $[\sin(t\vec{a}) + \vec{A}]$ is added to the traditional spiral equation, resulting in an adaptive and asymmetric motion pattern, as defined in Eq. 2:

$$\vec{X}(t+1) = \vec{D}' * e^{b * l} * \cos(2 * \pi * l) * (\sin(t * \vec{a}) + \vec{A}) + \vec{X}' \quad (2)$$

where $\vec{D}'$ represents the distance between the current agent and the best solution, b is a constant defining the spiral shape, l is a random number within $[-1, 1]$, and $\vec{A}$ and $\vec{a}$ are adaptive coefficients controlling convergence behavior.

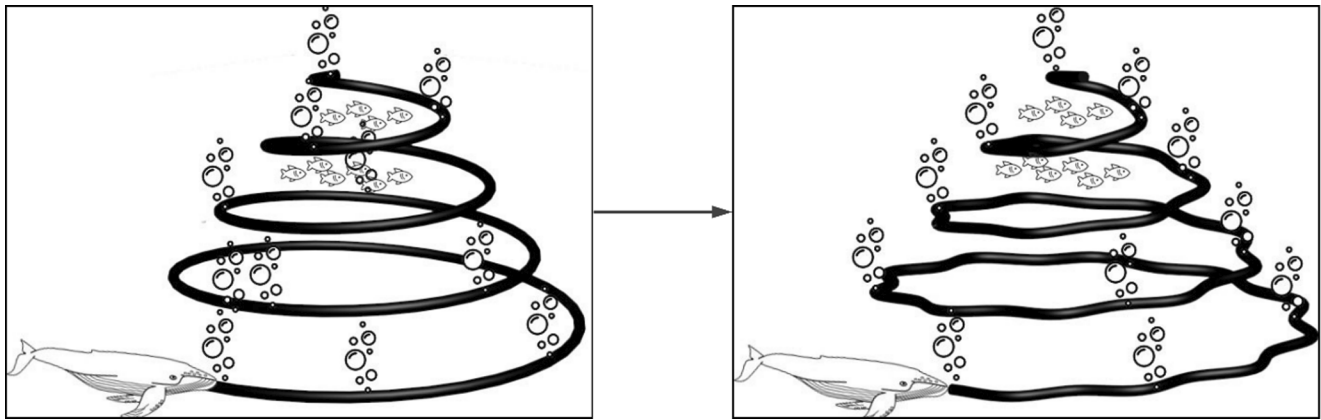


Fig. 2 a Standard and b improved whale optimization algorithms

This asymmetric sinusoidal trajectory allows agents to perform more flexible movements during both exploration and exploitation, improving convergence accuracy and avoiding stagnation.

The combination of population diversity monitoring and improved spiral motion enables IWOA to dynamically balance exploration and exploitation, maintain population diversity, and achieve faster and more accurate convergence than the standard WOA. A comparison of the standard and improved WOA behaviors is illustrated in Fig. 2a and b.

Deep Neural Network

DNNs have been widely used in various fields, including image processing, natural language processing, computer vision, and pattern recognition (Alzubaidi et al. 2021). In this study, the DNN architecture was used for automatic strawberry detection and ripeness classification, aiming to support intelligent decision-making in precision agriculture (Ibrahim 2021; Wang et al. 2022). The basic structure of

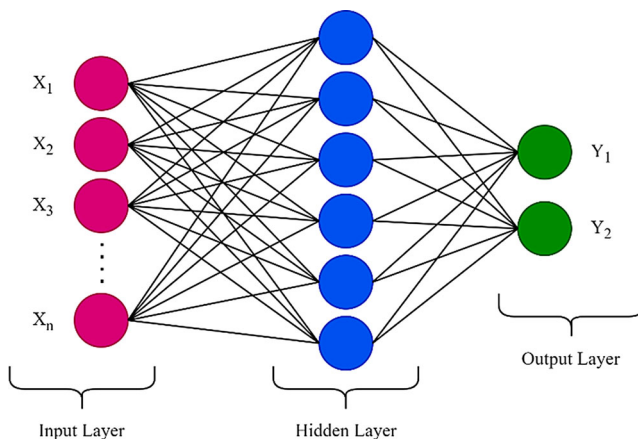


Fig. 3 The basic structure of a deep neural network

a DNN, shown in Fig. 3, consists of three main components: an input layer, one or more hidden layers, and an output layer. Each layer is composed of computational units known as perceptron (neuron), which process the incoming information and transmit it through weighted connections (Ghosh et al. 2024).

The network receives either raw input data or feature vectors derived from image preprocessing and computes the output of each perceptron according to Eq. 3 to determine the final decision of the network (Yuan et al. 2024).

$$net_j = \sum_{i=0}^n w_i x_i + b \quad (3)$$

where j represents the index of the perceptron whose output is being calculated, n denotes the number of input connections, w represents the connection weights, x corresponds to the input values, and b indicates the bias term. The computational process of a perceptron is illustrated in Fig. 4.

The activation function in the perceptron structure determines whether the neuron becomes active. Without an activation function, the network would behave like a sim-

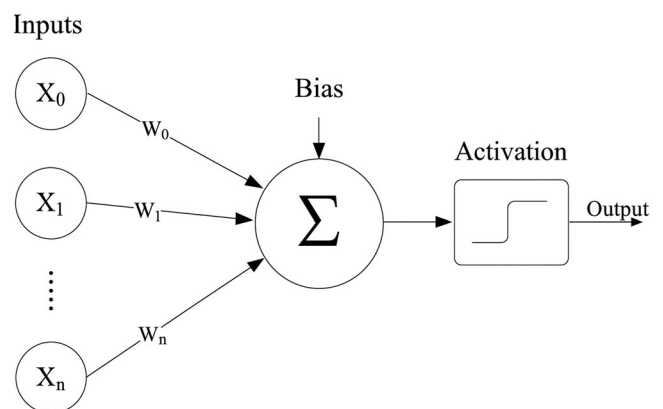


Fig. 4 The calculation of the perceptron

ple linear regression model with limited learning capability (Zhou et al. 2023; Locke et al. 2024). Commonly used activation functions in the literature include sigmoid, tanh, ReLU, leaky ReLU, and softmax (Montesinos López et al. 2022).

The hidden layers of a DNN play a crucial role in learning complex representations of the input data. Their associated weights are iteratively updated using either gradient-based or optimization-based techniques to minimize the output error (Yuan et al. 2024). The loss function quantifies the difference between the network's predicted output and the actual target value. Widely used loss functions include mean square error (MSE), mean absolute error (MAE), sum square error (SSE), binary cross-entropy (BCE), categorical cross-entropy (CCE), and root mean square error (RMSE) (Perminov et al. 2023). Based on the computed loss, the network's weights are adjusted iteratively until the desired accuracy is achieved or the predefined number of training iterations is reached. This process enables the DNN to progressively enhance its classification performance through error minimization. In this study, the DNN parameters were optimized using the IWOA as described in the section "Improved Whale Optimization Algorithm".

Autoencoder

An autoencoder is an unsupervised learning model widely used in deep learning frameworks to learn efficient data representations (Phadikar et al. 2023). Autoencoders are employed in various applications such as feature extraction (Chen et al. 2024), noise removal (Li and Zhao 2024), outlier detection (Du et al. 2022), encryption (Kumar et al. 2024), and data compression (Lungisani et al. 2024). The basic structure of an autoencoder designed for feature extraction is illustrated in Fig. 5.

As shown in Fig. 5, an autoencoder consists of two main components: an encoder and a decoder. In the encoder stage, the input image is compressed into a low-dimensional feature vector, which captures the most significant

and discriminative attributes of the input data. In the decoder stage, this feature vector is used to reconstruct the original image, allowing the network to learn meaningful representations by minimizing reconstruction error. During training, the network weights are optimized to minimize the difference between the reconstructed image and the original input, typically measured through a loss function such as MSE. Once the training process is completed, the encoder part of the network can be isolated and employed for feature extraction, providing a compact and information-rich representation of the strawberry texture features (Kumar et al. 2024).

Proposed IWOA-Based S-Det Framework

The proposed S-Det system integrates the IWOA with autoencoder and DNN architectures to achieve accurate and efficient strawberry ripeness classification. The overall workflow of the S-Det framework is illustrated in Fig. 6, and it consists of three consecutive stages: segmentation, feature extraction, and classification.

Segmentation

In the segmentation stage, the input strawberry image is divided into smaller texture patches of 32×32 pixels to isolate local visual regions that represent distinct ripeness characteristics (Fig. 7). This patch-based segmentation enables the system to focus on micro-level texture variations, such as color gradients, surface gloss, and seed distribution, which are strong indicators of ripeness. By analyzing these small, localized areas, the network gains a more detailed understanding of the strawberry's surface, thereby improving classification performance.

Feature Extraction

In the feature extraction stage, an IWOA-optimized autoencoder (IWOA-AUT) is employed to extract discriminative features from the segmented texture patches. The term IWOA-AUT refers to the autoencoder architecture whose parameters are optimized using the IWOA. The autoencoder consists of an input layer with M neurons, hidden layers with $(2M + 1)$ neurons, and an output layer with M neurons. The Softplus activation function is used in all layers to prevent inactive neurons, and the MSE between the input and reconstructed images is defined as the objective function (Eqs. 4 and 5).

$$F(x) = \log(1 + \exp(x)) \quad (4)$$

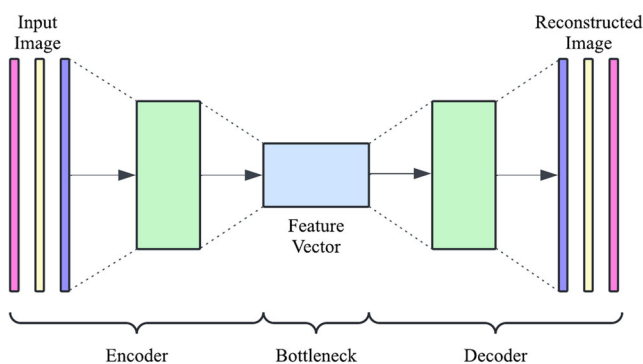


Fig. 5 The basic structure of an autoencoder

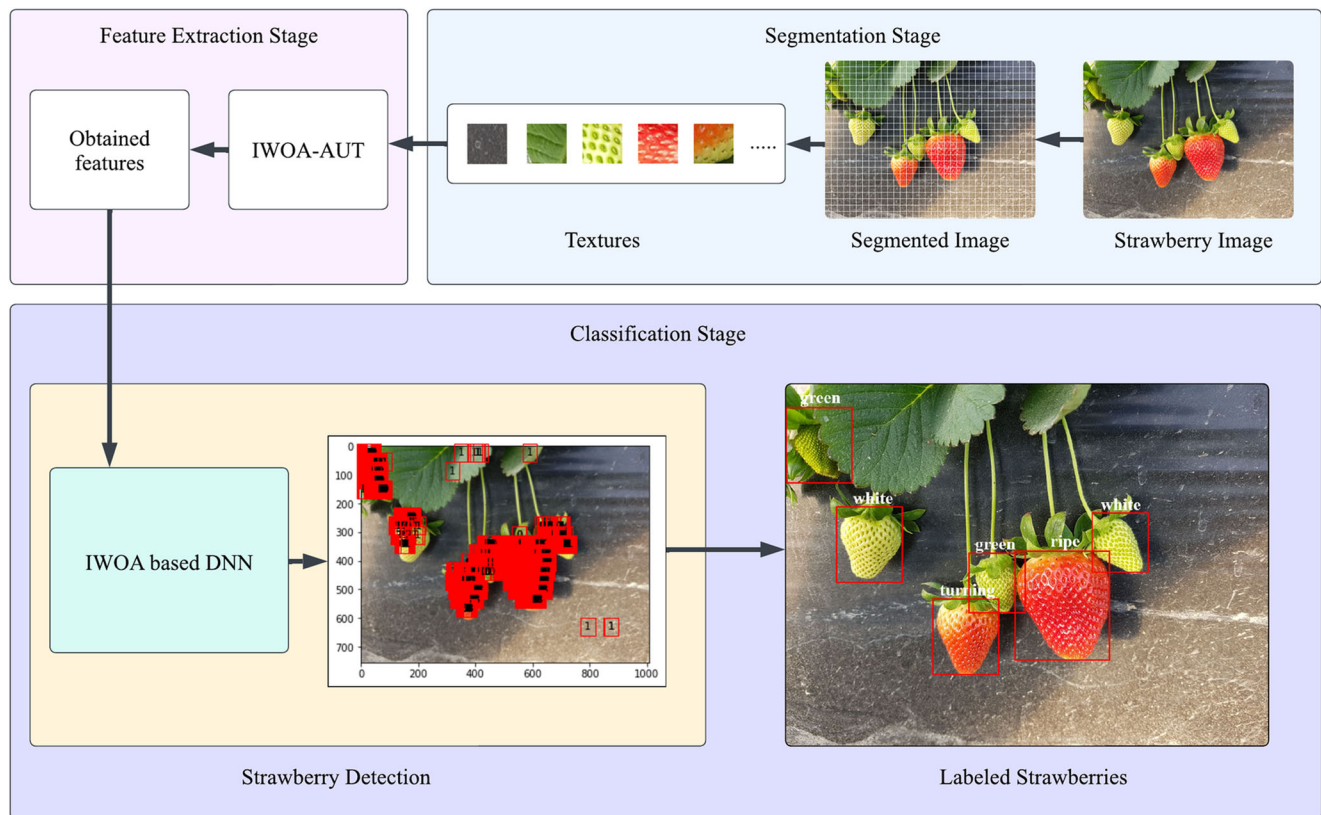
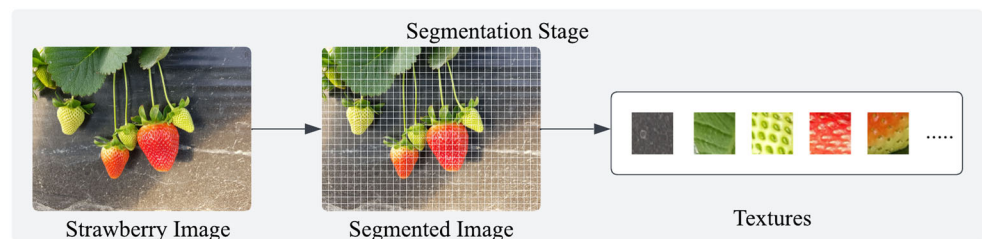


Fig. 6 Flowchart of the proposed strawberry detection and ripeness inspection system. *IWOA* Improved whale optimization algorithm, *DNN* Deep neural network

Fig. 7 The segmentation stage



$$MSE = \frac{\sum (y_i - y'_i)^2}{n} \quad (5)$$

To optimize the network parameters, the IWOA iteratively updates the autoencoder's weights by minimizing the MSE loss. This optimization enables the encoder to capture the most representative and non-redundant features from the strawberry textures. As a result, the IWOA-AUT produces low-dimensional, information-rich feature vectors that are robust against background noise and illumination variations.

Classification

In the final stage, the extracted feature vectors are classified using a DNN optimized by IWOA (IWOA-DNN). The DNN architecture comprises an input layer corresponding to the length of the feature vector (N), a hidden layer with $(2N + 1)$ neurons, and an output layer with five neurons, each representing one ripeness category: white, green, turning, ripe, and background regions. The Softmax activation function is used to produce normalized class probabilities for each output node.

IWOA optimizes the DNN weights to minimize the classification loss function, allowing the model to avoid local minima and achieve faster convergence. This optimization

process enhances classification accuracy and stability compared to conventional gradient-based methods.

Finally, the classified strawberries are labeled in the output image according to the density of the related class (d_i), defined in Eq. 6:

$$d_i = \frac{m_i}{v} \quad (6)$$

where i represents the number of classes, m represents the number of detected textures for a specific class, and v denotes the total number of textures in the image. This density measure allows the model to quantify the dominant ripeness stage of each strawberry, resulting in a clear and interpretable output.

Results and Discussion

To evaluate the efficiency and robustness of the proposed IWOA-based S-Det framework, extensive experiments were conducted on the StrawTex dataset, along with comparative analyses using classical machine learning models, optimization-based DNNs, and pre-trained deep learning architectures. This section presents the experimental setup, evaluation metrics, and detailed performance comparisons, followed by an in-depth discussion of the results.

Performance Evaluation Metrics

Performance evaluation metrics are essential for quantitatively assessing the effectiveness of classification models. In this study, six widely used metrics were employed: Accuracy (Acc), Precision (Pre), Recall (Rec), False Positive Rate (FPR), False Negative Rate (FNR), and F1-score ($F1$) (Toprak 2024). Accuracy represents the ratio of correctly classified samples to the total number of samples. Precision indicates the proportion of true positive predictions among all positive predictions, while Recall measures the proportion of true positives among all actual positive cases. The False Positive Rate and False Negative Rate quantify incorrect classifications for negative and positive cases, respectively. The F1-score, defined as the harmonic mean of Precision and Recall, provides a balanced measure of classification performance, particularly useful in multi-class problems. The mathematical definitions of these metrics are presented in Eqs. 7–10 where TP , TN , FP , and FN denote the counts of true positive, true negative, false positive, and false negative instances, respectively:

$$Acc = \frac{TP + TN}{TP + TN + FP + FN} \quad (7)$$

$$Pre = \frac{TP}{TP + FP} \quad (8)$$

$$Rec = \frac{TP}{TP + FN} \quad (9)$$

$$F1 = 2 \times \frac{Pre \times Rec}{Pre + Rec} \quad (10)$$

Comparative Results

The comparative results of 12 classification models on the StrawTex dataset are presented in Table 1.

The proposed IWOA-DNN outperformed all baseline methods, achieving the highest accuracy (98.23%), precision (98.25%), recall (98.26%), and F1-score (0.9824), while maintaining zero false positives ($FPR=0.0$) and a minimal false negative rate ($FNR=0.0008$).

Models trained with metaheuristic optimization—artificial bee colony-optimized DNN (ABC-DNN), genetic algorithm-optimized DNN (GA-DNN), particle swarm optimization-optimized DNN (PSO-DNN), whale optimization algorithm-optimized DNN (WOA-DNN, improved whale optimization algorithm-optimized DNN (IWOA-DNN)—and pre-trained networks (VGG16, DenseNet201, ResNet50) demonstrated superior performance compared to traditional classifiers— k -nearest neighbors (KNN), support vector machine (SVM), random forest (RF). Among all tested algorithms, the proposed IWOA-DNN achieved the top ranking (Rank 1) in the Friedman ranking test. KNN, SVM, RF, BP-DNN, ABC-DNN, GA-DNN, PSO-DNN, and WOA-DNN algorithms were tested using 32×32 input images. In contrast, VGG16, DenseNet201, and ResNet50 were trained on 224×224 images (Dede et al. 2025). To ensure a fair comparison and avoid an unfair advantage, the 32×32 input images were upscaled to 224×224 using the nearest-neighbor interpolation method, which assigns each new pixel the value of the closest pixel in the original image without blending or smoothing (Song et al. 2025). The experimental study was performed using image data split as 90% training and 10% testing, with 10-fold cross-validation. The fact that the test results given in Table 1 were obtained through 10-fold cross-validation indicates that the results are not due to random chance but rather reflect a consistent and reliable performance across different data splits.

The convergence behavior of the proposed IWOA-DNN model demonstrates stable learning performance and the absence of overfitting or underfitting. Compared to other optimization algorithms, IWOA exhibited faster convergence and smoother error reduction due to its enhanced spiral motion and diversity maintenance strategies.

Table 1 Test results of deep learning models trained with optimization algorithms, classical classifiers, and pre-trained deep learning models for the StrawTex dataset

Classifier	Parameters	Performance evaluation metrics						
		ACC	PRE	REC	FPR	FNR	F1	Rank
KNN	k: 3	0.9217	0.9235	0.9217	0.0196	0.0786	0.9215	11
SVM	Kernel: linear	0.9206	0.9213	0.9206	0.0199	0.0788	0.9206	12
RF	Estimators: 300	0.9412	0.9420	0.9412	0.0146	0.0587	0.9411	10
BP-DNN	Epoch: 250, Activation: Softplus, Learning Rate: 0.001, Loss: SCCE, Optimizer: Adam	0.9506	0.9515	0.9506	0.0123	0.0497	0.9504	9
ABC-DNN	Epoch: 250, Population: 10, n_limits:5	0.9633	0.9644	0.9641	0.0149	0.0103	0.9635	7
GA-DNN	Epoch: 250, Population: 10, Selection: Tournament, Crossover Probability: 0.9, Mutation Probability: 0.05	0.9636	0.9644	0.9640	0.0142	0.0095	0.9635	6
PSO-DNN	Epoch: 250, Population: 10, c1: 2.1, c2: 2.1, w_min: 0.4, w_max: 0.9	0.9762	0.9766	0.9764	0.0	0.0017	0.9762	4
WOA-DNN	Epoch: 250, Population: 10	0.9772	0.9774	0.9773	0.0	0.0012	0.9771	3
VGG16	Epoch: 250, Activation: Sigmoid, Learning Rate: 0.001, Weights: ImageNet, Loss: SCCE, Optimizer: Adam	0.9528	0.9351	0.9409	0.0149	0.0601	0.9380	8
DenseNet201	Epoch: 250, Activation: Softplus, Learning Rate: 0.001, Weights: ImageNet, Loss: SCCE, Optimizer: Adam	0.9757	0.9778	0.9757	0.0061	0.0243	0.9749	5
Resnet50	Epoch: 250, Activation: Softplus, Learning Rate: 0.001, Weights: ImageNet, Loss: SCCE, Optimizer: Adam	0.9773	0.9793	0.9773	0.0057	0.0225	0.9770	2
IWOA-DNN	Epoch: 250, Population: 10, Cosine similarity: 0.9	0.9823	0.9825	0.9826	0.0	0.0008	0.9824	1

ACC accuracy, PRE precision, REC recall, FPR false positive rate, FNR false negative rate, F1 F1-score, KNN k-nearest neighbors, SVM support vector machine, RF random forest, DNN deep neural network, BP-DNN backpropagation-based deep neural network, ABC-DNN artificial bee colony-optimized deep neural network, GA-DNN genetic algorithm-optimized deep neural network, PSO-DNN particle swarm optimization-optimized deep neural network, WOA-DNN whale optimization algorithm-optimized deep neural network, IWOA-DNN improved whale optimization algorithm-optimized deep neural network

The confusion matrix of the proposed IWOA-DNN model on the StrawTex dataset is presented in Fig. 8. The diagonal elements represent correctly classified samples for each ripeness category, while the off-diagonal elements indicate misclassifications. The matrix clearly shows that the proposed model achieves near-perfect classification across all five classes, with only a few misclassifications between the *turning* and *ripe* categories due to their visual similarity in texture and color. Overall, the confusion matrix analysis validates that the IWOA-DNN achieved highly reliable recognition across all ripeness categories, confirming the discriminative capability of the IWOA-AUT features and the robustness of the IWOA-optimized DNN in distinguishing subtle ripeness differences.

The confusion matrix, shown in Fig. 8, demonstrates that the proposed IWOA-DNN model achieved highly reliable classification performance across all ripeness categories, with over 98% correct predictions and only minimal confusion observed between visually similar stages such as *turning* and *ripe*.

Ablation Study on the Contribution of IWOA-AE and IWOA-DNN

In this study, an ablation analysis was conducted to determine the contribution of each component to the success of the proposed S-Det system. Specifically, the effects of the IWOA on both the autoencoder used in the feature extrac-

tion stage (IWOA-AE) and the deep neural network used in the classification stage (IWOA-DNN) were analyzed separately. In this context, four different model variants were compared using the 10-fold cross-validation method:

- Variant 1 (IWOA-AE + IWOA-DNN): The proposed full model
- Variant 2 (IWOA-AE + Normal DNN): Model without IWOA only in the classification stage
- Variant 3 (Normal AE + IWOA-DNN): Model using a classical autoencoder only in the feature extraction stage
- Variant 4 (Normal AE + Normal DNN): Baseline control model without IWOA in any stage

Disabling the IWOA-DNN component (Variant 2) caused an approximate 2.52% decrease in the average test accuracy. Disabling the IWOA-AE component (Variant 3) led to a more pronounced decrease of approximately 4.14% in the average test accuracy. When both IWOA components were removed together (Variant 4), the total accuracy loss was observed to be approximately 7.96%. These findings clearly demonstrate that the success of the proposed system stems from the combined use of both components. However, it is observed that the IWOA-optimized autoencoder provides a higher individual contribution compared to the classifier optimization. This indicates that obtaining discriminative and low-dimensional features has a critical impact on classification performance. In conclusion, the

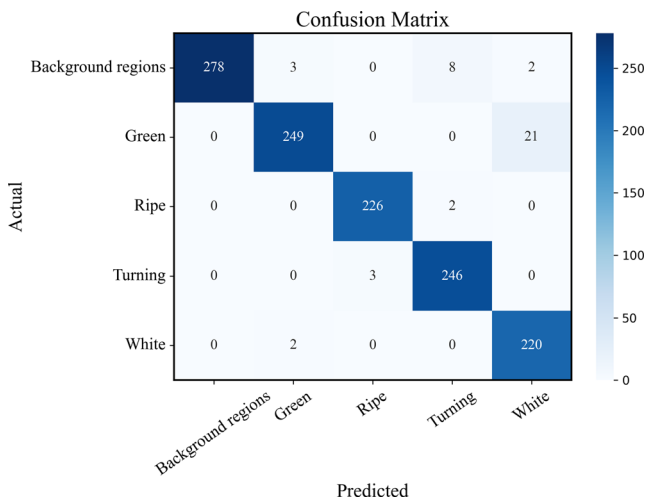


Fig. 8 Confusion matrix of the proposed improved whale optimization algorithm-optimized deep neural network model on the StrawTex dataset

conducted ablation study quantitatively and experimentally validates the contributions of the IWOA-AUT and IWOA-DNN components to the success of the proposed S-Det system; it reveals that the system's superior performance arises from the balanced and joint utilization of these two components.

Discussion

The superior performance of the proposed IWOA-based S-Det framework can be attributed to several factors:

- **Enhanced Optimization via IWOA:** The integration of population diversity monitoring prevented premature convergence by maintaining exploration in later iterations. The improved spiral update mechanism contributed to more precise local search, accelerating convergence toward the global optimum.
- **Feature Representation through IWOA-AUT:** The IWOA-optimized autoencoder effectively extracted low-dimensional, noise-invariant features from strawberry texture patches. These compact representations improved the discriminative capacity of the classifier, especially under varying illumination and background conditions.
- **High Classification Stability:** The combination of IWOA-AUT and IWOA-DNN achieved the best trade-off between accuracy and computational cost, outperforming heavy pre-trained networks such as ResNet50 and DenseNet201 while requiring fewer parameters and training resources.
- **Real-World Applicability:** The robustness of the proposed framework demonstrates its potential for real-time implementation in agricultural automation systems, such as smart sorting lines or robotic harvesting units.

- **Confusion Matrix Validation:** The confusion matrix analysis further confirmed that the proposed IWOA-DNN achieved highly reliable recognition across all ripeness categories, with minimal confusion between visually similar classes such as *turning* and *ripe*, reinforcing its robustness under varying field conditions.

Overall, the IWOA-based S-Det framework effectively addresses the limitations of gradient-based optimization and standard metaheuristic methods by offering improved convergence dynamics, enhanced feature discrimination, and superior classification accuracy.

Conclusion

Accurate determination of strawberry ripeness levels is essential for optimizing harvest timing, maintaining fruit quality, and reducing postharvest losses in precision agriculture. In this study, an improved whale optimization algorithm (IWOA)-based strawberry detection and ripeness inspection (S-Det) system was developed to address these challenges through an intelligent and automated classification framework. The proposed system integrates an IWOA-optimized autoencoder (IWOA-AUT) for effective feature extraction with an IWOA-optimized deep neural network (IWOA-DNN) for high-accuracy classification, ensuring a balanced exploration-exploitation process and avoiding premature convergence.

The proposed model achieved an overall accuracy of 98.23% and F1-score of 0.9824, confirming its effectiveness for strawberry ripeness detection under realistic field conditions. Experimental evaluations on the StrawTex dataset showed that the proposed IWOA-DNN achieved superior performance compared to both traditional and state-of-the-art models. Specifically, the S-Det system reached an accuracy of 98.23%, precision of 98.25%, recall of 98.26%, and F1-score of 0.9824, surpassing classical classifiers such as *k*-nearest neighbors (92.17%), support vector machine (92.06%), and random forest (94.12%), as well as optimization-based networks like particle swarm optimization-optimized DNN (97.62%) and whale optimization algorithm-optimized DNN (97.72%). Moreover, the proposed method outperformed pre-trained deep learning architectures, including DenseNet201 (97.57%) and ResNet50 (97.73%), demonstrating faster convergence and improved generalization.

These results confirm the robustness and efficiency of the proposed IWOA in optimizing deep learning models for agricultural image analysis. Overall, the IWOA-based S-Det system offers a powerful and adaptive framework for fruit ripeness detection, contributing to automation and precision in modern agriculture. The proposed method's high

accuracy and computational efficiency make it suitable for real-time implementation in sorting and robotic harvesting systems. In future studies, this approach can be extended to other fruit and vegetable species to support the development of intelligent, data-driven decision-making in smart farming applications.

Author Contribution All authors contributed equally to this work.

Data Availability The data that supports the findings of this study are available from the corresponding author upon reasonable request.

Conflict of interest Ö. Özcan, M. Karaaltun and M. Hacıbeyoğlu declare that they have no competing interests.

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